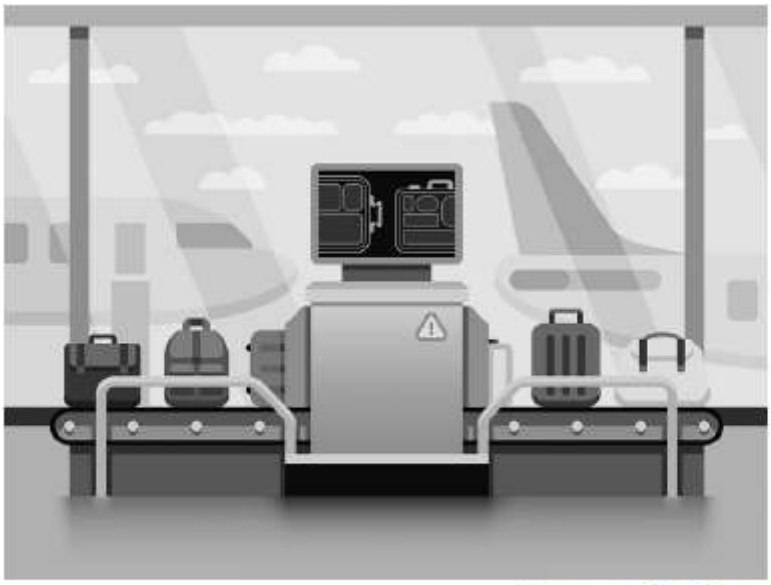
Questions

Q1.

This question is about waves in the electromagnetic (e-m) spectrum.

X-rays are also part of the e-m spectrum.

Figure 2 shows an airport security scanner using X-rays to scan passengers' bags.



(Source: © Net Vector / Shutterstock)

Figure 2

(i) Explain why X-rays are used to scan passengers' bags.

(2)

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(ii) Explain why passengers are **not** scanned with X-rays.

(2)

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(Total for question = 4 marks)

Q2.

This question is about light.

* Figure 11a shows refraction of light at a boundary between glass and air.
Figure 11b shows total internal reflection of light at a boundary between glass and air.

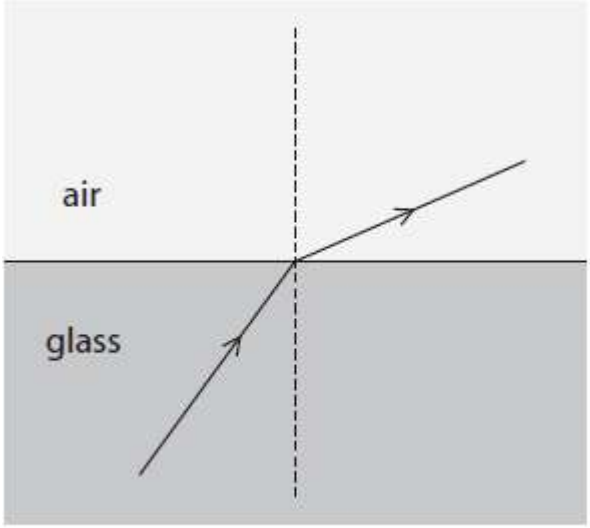

Figure 11a

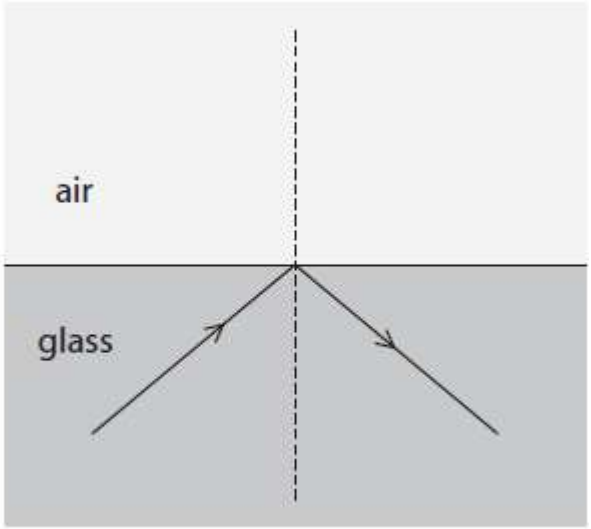

Figure 11b

Use Figure 11a and Figure 11b to explain refraction and total internal reflection.

You may add to Figure 11a and Figure 11b to help with your answer.

(6)

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(Total for question = 6 marks)

Q3.

Some television remote controls use infrared radiation and other remote controls use radio waves.

Explain why an infrared remote control may not switch on the television from behind an armchair but a radio wave remote control always will.

(2)

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(Total for question = 2 marks)

Q4.

Equal volumes of hot water are added to two cans.
The cans are identical apart from their surfaces.
One can has a black surface and the other can has a silver surface.

The cans are left to cool and their temperatures are monitored.

The graph in Figure 6 shows the results.

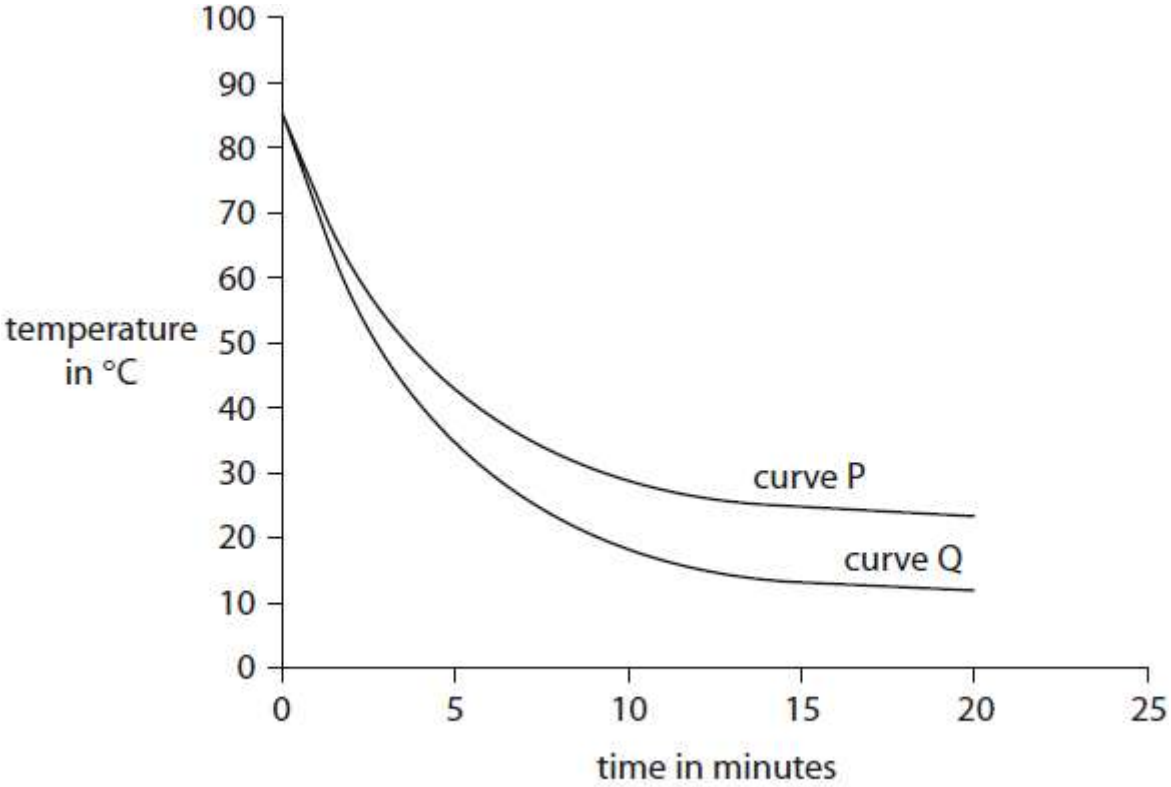

Figure 6

Explain, using evidence from the graph, which curve is for the black can and which curve is for the silver can.

(2)

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(Total for question = 2 marks)

Q5.

Figure 7 shows light reflecting from two mirrors in a bathroom.

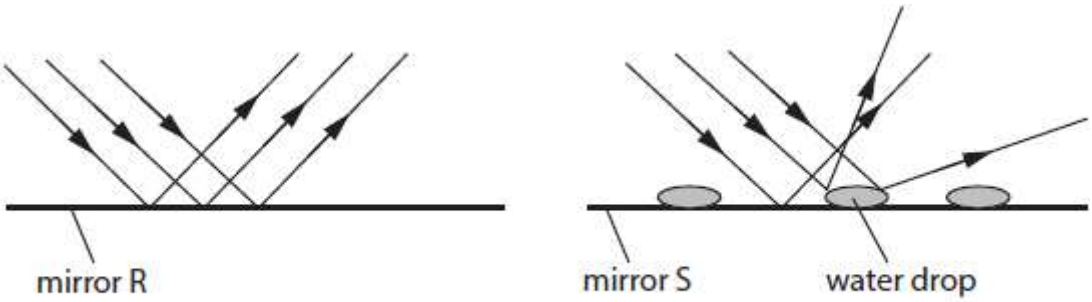

Figure 7

Mirror R has a dry surface.

Mirror S has lots of small drops of water on the surface.

A person looks into each mirror.

Explain the difference between the image seen in mirror R and the image seen in mirror S.

(2)

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(Total for question = 2 marks)

Q6.

A person warms their hands in front of a hot fire as shown in Figure 2.

(Source: © Andreas Saldavs/Shutterstock)

Figure 2

Use words from the box to complete the following sentences.

chemical	infrared	radio	thermal	ultraviolet

(2)

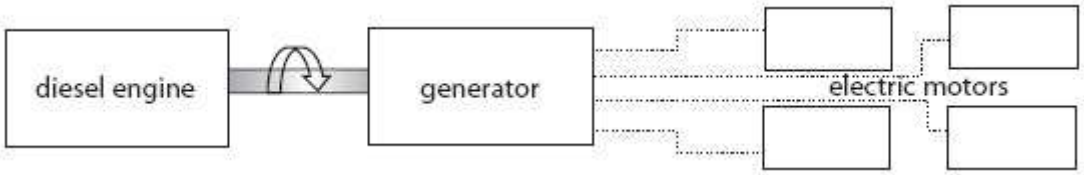
The electromagnetic waves that the fire mostly emits are ..... waves.

These waves transfer ..... energy to the hands.

(Total for question = 2 marks)

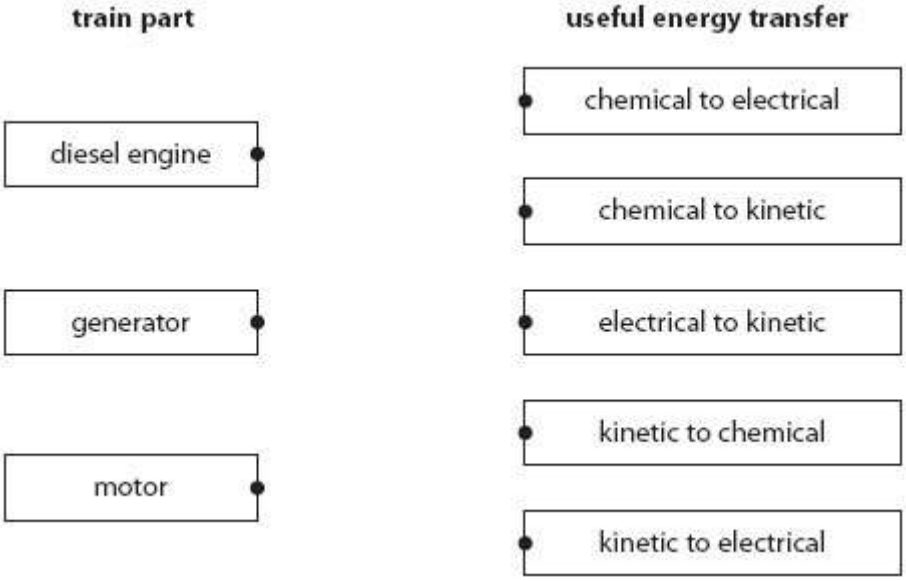
Q7.

(a) A train is powered by a diesel engine.
The diesel engine is used to turn a generator.
The generator provides electricity for electric motors which drive the wheels.



(i) Draw one straight line from each train part to its useful energy transfer.

(3)



(ii) State **one** example of a non-useful energy transfer in the motor.

(1)

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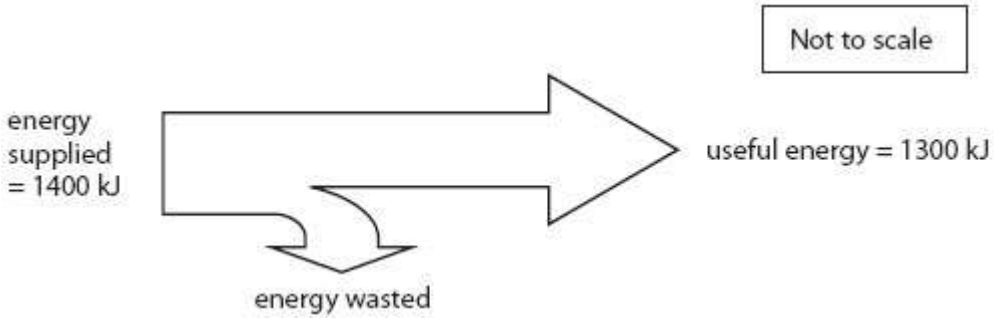
(b) The diagram represents the energy transfer in one second in the generator.



(i) Calculate the amount of energy wasted in one second in the generator.

(1)

.....

(ii) Calculate the efficiency of the generator.

(2)

.....

(c) The electric motors which drive the wheels are painted black.
Suggest why the motors are painted black.

(1)

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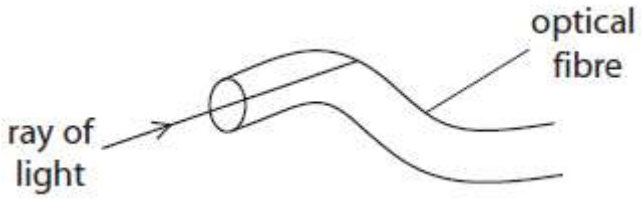
(Total for Question is 8 marks)

Q8.

(a) An endoscope uses glass fibres to enable doctors to 'see' inside a person.



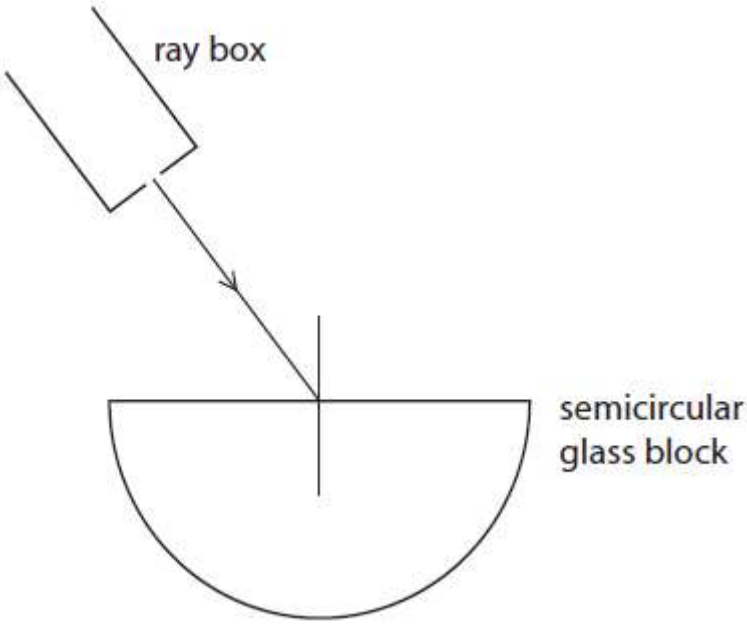
Complete the sentence by putting a cross (☒) in the box next to your answer.
Rays of light travel along the glass fibres by

(1)

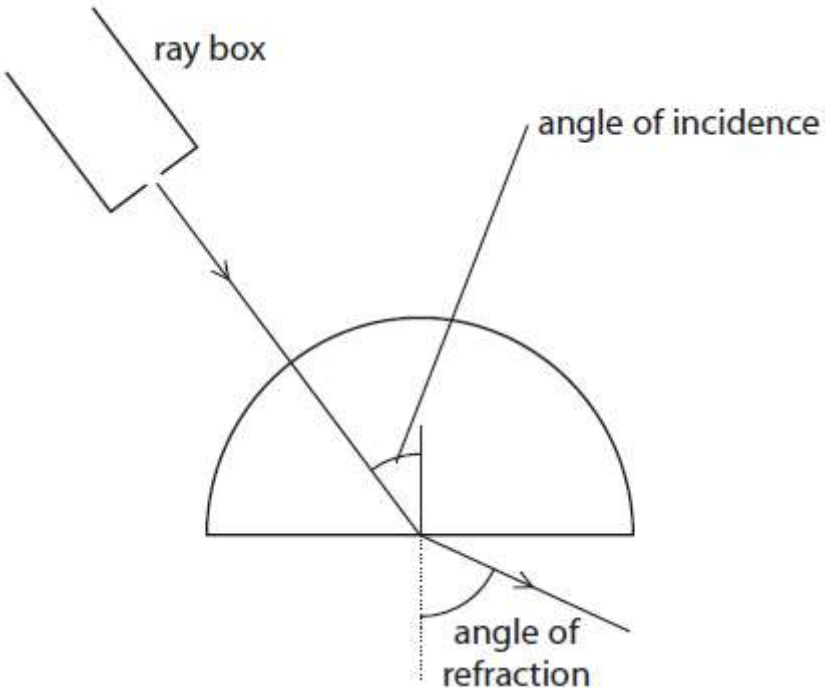
☐ **A** total internal correction
☐ **B** total internal diffraction
☐ **C** total internal reflection
☐ **D** total internal refraction

(b) The diagram shows a ray box shining a ray of light at a semicircular glass block.
Complete the diagram to show the path of the ray of light inside the glass block.

(2)

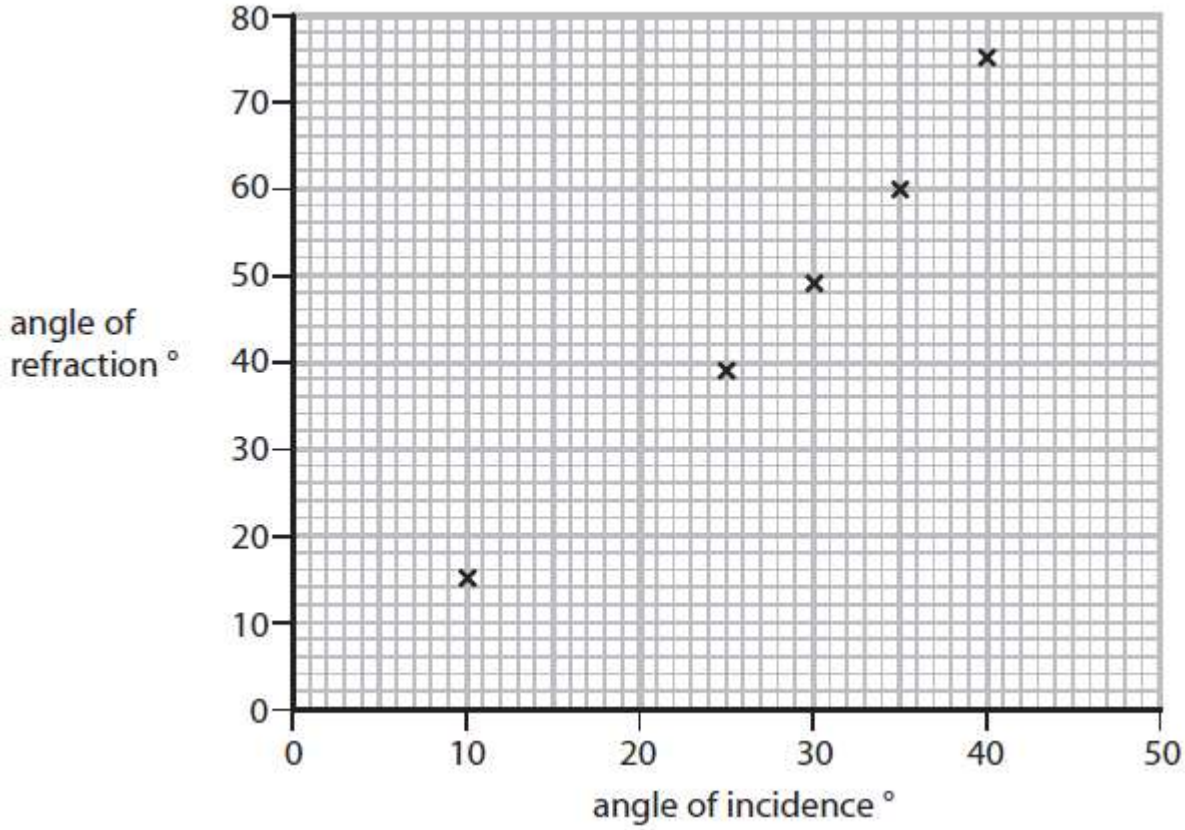


(c) A student shines a ray of light into the block as shown.



He measures the angle of incidence inside the block as shown.
He measures the angle of refraction outside the block as shown.
He repeats these measurements for different angles of incidence.
He records his results in this table and plots a graph.

angle of incidence °	10	15	20	25	30	35	40
angle of refraction °	15	23	31	39	49	60	75



(i) Two points are highlighted in the table.
Plot these points on the graph.

(2)

(ii) Draw the curve of best fit.

(1)

*(iii) The student continues to increase the angle of incidence until it reaches 80°.
The critical angle for this glass is 42°.
Explain what happens to the ray of light as the angle of incidence is increased from 10° to 80°.

(6)

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Q9.

Some students repeat Herschel's experiment.

(a) They place four identical thermometers, **P**, **Q**, **R** and **S**, in the shade.

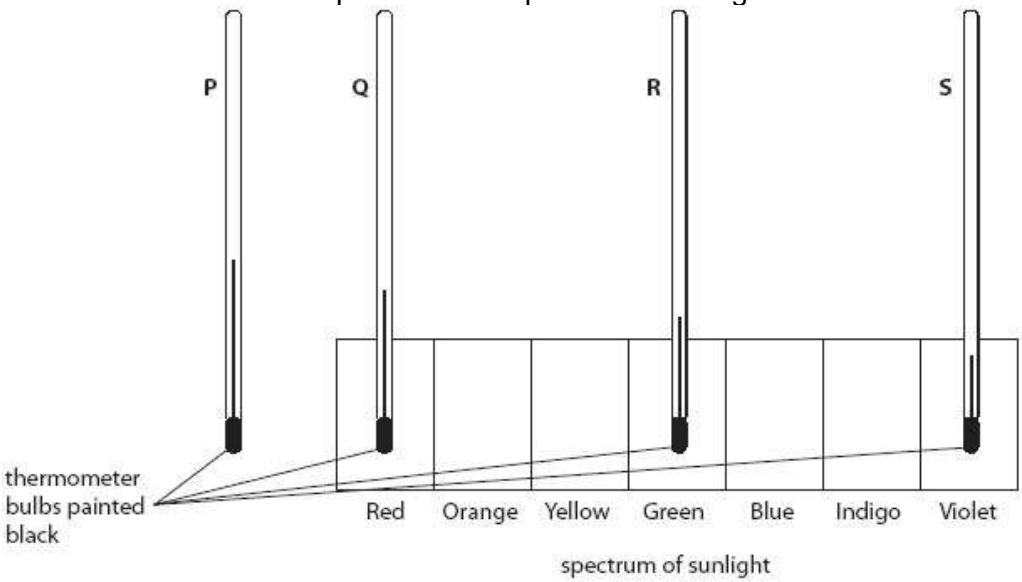
The table shows some of the readings on the thermometers in the shade.

Complete the table to show the reading on thermometer **Q**.

(1)

	thermometers in the shade			
	P	**Q**	**R**	**S**
temperature / °C	18		18	18

(b) Then the four thermometers are placed in the spectrum of sunlight as shown.



(i) The bulbs of the thermometers are painted black.

State why the bulbs of the thermometers are painted black.

(1)

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.....

(ii) The table shows the readings on the thermometers in the spectrum of sunlight.

	thermometers in the spectrum of sunlight			
	P	**Q**	**R**	**S**
temperature / °C	25	23	21	20

Describe conclusions that can be made from the students' results.

(2)

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(iii) State why the students put the four thermometers together in the shade before placing them in the spectrum

of sunlight.

(1)

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(c) Another scientist, Ritter, discovered ultraviolet radiation.

(i) Complete the sentence by putting a cross (■) in the box next to your answer.

Too much exposure to ultraviolet radiation may cause

(1)

☐ **A** deafness

☐ **B** heating of internal body cells

☐ **C** damage to the eyes

☐ **D** damage to the bone cells

(ii) Three signals, ultraviolet, visible light and infrared, are sent from the surface of the Moon to an orbiting spacecraft.

The three signals are sent at the same time.

Which of these is correct for the signals arriving at the spacecraft?

Put a cross (■) in the box next to your answer.

(1)

☐ **A** the visible light signal arrives first

☐ **B** the ultraviolet signal arrives first

☐ **C** the infrared signal arrives first

☐ **D** all three signals arrive at the same time

(iii) Describe **one** use of ultraviolet radiation.

(2)

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(Total for Question is 9 marks)

Q10.

The electric motors which drive the wheels are painted black.
Suggest why the motors are painted black.

(1)

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Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	an explanation linking (X-rays/they) pass through/penetrate (the bags/cases) (1) to see contents/to show objects of greater density (1)	 accept see through accept look/see inside accept check contents/dangerous items	(2) AO1
Question Number	**Answer**	**Additional guidance**	**Mark**
(ii)	an explanation linking X-rays/they are ionising (1) cause cancer/mutations (of cells/DNA) (1)	 accept harmful/dangerous accept a description of ionising accept high energy accept kill/damage cells	(2) AO2

Q2.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **Refraction** • Angle of incidence marked • Angle of refraction marked • Angles are measured from the normal • Angle of refraction is bigger than the angle of incidence • Rays of light travel in straight lines • Refraction occurs at a boundary between two materials of different (optical) density • The angle of incidence is less than the angle of refraction when light passes into a less dense medium (glass into air) • Refraction is a change in direction of a light ray. • Refracted rays bend away from the normal when light passes into a less dense medium (glass into air) • The ray in the more dense medium (glass) travels more slowly ORA **Total Internal Reflection** • Possible critical angle marked • Light stays inside the glass • Only occurs when the incident light is in the more dense medium • Only occurs when the incident angle is equal to greater than the critical angle • Critical angle for glass is about 42° • Angle of incidence is equal to the angle of reflection	**(6)** **AO1**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3–4	Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5–6	Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated fact(s) about refraction or total internal reflection(TIR)	Possible candidate responses naming of any rays of light or any angles in text or on diagrams light changes direction/bends TIR ray stays inside the glass / does not go into air refracted ray goes through glass and air
Level 2	3–4	Additional guidance simple description of refraction and TIR or facts about one and more detail of the other	Possible candidate responses Angle or ray identified For refraction light changes direction from glass into air or TIR angles are equal inside the glass
Level 3	5–6	Additional guidance detailed description of refraction and TIR	Possible candidate responses For refraction light changes direction from glass into air AND TIR angles are equal inside the glass

Q3.

Question Number	Answer	Additional guidance	Mark
	an explanation linking: infrared is absorbed / blocked (by the armchair/objects) / cannot pass through it **OR** radio waves can go through (the armchair/objects) (1) **WITH** (infrared and radio have) different wavelengths / frequencies OR infrared requires 'line-of-sight' (idea) OR radio waves do not require 'line-of-sight' (idea) OR diffraction (idea) (1)	allow stopped transmitted accept comparison	**(2)**

Q4.

Question Number	Answer	Additional guidance	Mark
	an explanation linking **two** from: identifies curve Q is the black can OR that curve P is for the silver can (1) (because, as) black is a better emitter (then) (curve) Q shows a faster rate of cooling (1) OR (curve) Q is steeper (1)	 IGNORE reference to absorb allow Q is at a lower temperature/stays cooler OR P is at a higher temperature /stays hotter may quote numbers from graph e.g. curve Q cools by 75(°C) in 20 minutes but curve P only shows a 60(°C) in the same time	**(2)**

Q5.

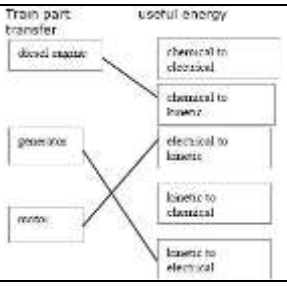
Question number	Answer	Additional guidance	Mark
	an explanation linking blurred / fuzzy/ distorted (1) rays are reflected in different directions / randomly (from mirror S) (1)	allow reverse arguments (reflection in mirror) S is diffuse allow light for rays (reflection in mirror) R is specular rays reflected from R are parallel	**2** **AO3.2**

Q6.

Question number	Answer	Additional guidance	Mark
	infrared (1) thermal (1)	must be in first sentence space must be in second sentence space award 2 marks for answers in this **order**	**(2)** **AO2**

Q7.

	Answer	Acceptable answers	Mark
(ai)	A line connecting a train part with a useful energy transfer as shown below (1)	Lines need not be straight Ignore any arrow heads drawn Note: if more than one line is drawn from a train	**(3)**

		part then zero mark for that train part.	
(aii)	(transfer of energy to) thermal (1)	heat/sound	**(1)**
(bi)	1400 – 1300 (= 100) (kJ) (1)		**(1)**
(bii)	Substitution (1) 1300 / 1400 × 100 Evaluation (1) 93(%) or 0.93	A value which rounds to 93(%) or 0.93 Correct answer with no working scores 2 marks	**(2)**
(c)	Any one from black is a good thermal radiator (1) (helps to) prevent motors overheating (1)	(good) emitter (helps to) remove wasted energy/ heat (from the motor)	**(1)**

Q8.

Question Number	Answer	Acceptable answers	Mark
(a)	**C** total internal reflection		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	drawn ray changing direction at the boundary as it goes into the glass (1) towards the normal after entering the glass (1)	line should be near to straight allow dotted lines / don't need arrows ignore lines outside the glass block **reject** lines in the block going to the left of the normal.	**(2)**

Question Number	Answer	Acceptable answers	Mark
(c) (i)	each point plotted to within half a small square.	i.e. first plotted point to lie within 22→24, second 30→32 must be plotted; can't assume under any line they draw	**(2)**

Question Number	Answer	Acceptable answers	Mark
(c)(ii)	one best fit **curved** line close to / through most of points	**the best fit line does not have to extend beyond the plotted points.** reject very shaky / point to point drawing and tramlining (multiple lines). Reject lines forced through the origin involving kinks / changing directions in order to do so.	**(1)**

Question Number		Indicative Content	Mark
QWC	**(c)(iii)**	An explanation including some of the following ideas :- • idea(s) of refraction / reflection / TIR conveyed • as angle of incidence increases so angle of refraction increases • becomes increasingly steep/ gradient increases (non-linear) • because of refraction (light speeding up) • at 42° light stops refracting / up to 42° limit idea • critical angle case described in terms of an emerging ray along the boundary (or similar) • beyond 42° reflection occurs • this is total internal reflection • then angle of incidence = angle of reflection (in the glass)	**(6)**
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited explanation including isolated facts e.g. about light bending / refraction • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple explanation linking facts about a part of the process e.g. as angle of incidence increases so angle of refraction increases • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed explanation which links ideas about more than one part of the whole process e.g. an accurate explanation of refraction before 42° and TIR after 42° • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q9.

	Answer	**Acceptable answers**	**Mark**
(a)	18 (°C) (1) Ignore any unit given by candidate.		**(1)**
(b)(i)	(black is the) best absorber (of radiation/heat)(compared to other	(Black/it) absorbs more (Black/it) is a good absorber (of radiation/heat) (black is) good at taking in	**(1)**

	colours) (1)	(radiation/heat) Ignore (black is a) good emitter Ignore light. Reject black attracts heat/radiation	
(b)(ii)	• Heating effect/temperature greatest at/beyond red (1) • (There is) radiation beyond red (end of spectrum)(1)	(idea of) different colours have different heating effects (Radiation from) sunlight causes a heating effect accept reverse argument Infrared/IR (beyond red end of spectrum)	**(2)**
(b)(iii)	Any **one** from • To check the thermometers produce the same readings/ temperature (under the same conditions) (1) • To show that temperature changes.	(To check) they were all at the same temperature (before starting the experiment.) To be able to make a comparison (between shade and light) (To allow them to carry out a) fair test	**(1)**
(c)(i)	C damage to the eyes (1)		**(3)**
(c)(ii)	D all three signals arrive at the same time (1)		**(1)**
(c)iii	Description linking **one** of the following pairs: • security marking (1) • ink absorbs UV and re-radiates (visible) light (1) • fluorescent lamps (1) • coating absorbs UV and reradiates (visible) light (1) • genuine bank notes (1) • watermark absorbs UV and reradiates (visible) light (1) • disinfecting water (1) • UV kills bacteria (1) • sun beds (1)	invisible ink/smart water glows under UV (outside of) lamp glows when hit by UV forgeries/fake bank notes/passports/fingerprints/ body fluids etc markings glow under UV tanning beds tans the skin /the body e.g. disco lighting (1) makes clothing glow (1)	**(2)**

	• UV absorbed by (melanin in) skin (1) Any suitable use gains 1 mark Any suitable use + detail gains 2 marks		

(Total for question = 9 marks)

Q10.

	Answer	Acceptable answers	Mark
	Any one from black is a good thermal radiator (1) (helps to) prevent motors overheating (1)	(good) emitter (helps to) remove wasted energy/ heat (from the motor)	**(1)**